KAIZEN MATHS

Lesson starter

Indices and Exponent Laws

Fundamentals of Index Notation

Name _____

S1

Write in index form: $3 \times 3 \times 3$

S2

Evaluate: 4^2

S3

Express 24 as a product of prime factors in index form

S4

Simplify: a^{-1}

Show your working

KAIZEN MATHS

Lesson starter

Indices and Exponent Laws

Fundamentals of Index Notation

Name _____

S1

Write in index form: $3 \times 3 \times 3$

S2

Evaluate: 4^2

S3

Express 24 as a product of prime factors in index form

S4

Simplify: a^{-1}

ALG-934A-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Indices and Exponent Laws

Fundamentals of Index Notation

Name _____

E1

Write in index form: $5 \times 5 \times 5 \times 5 \times 5$

E2

Evaluate: 4^3

E3

Express 48 as a product of prime factors in index form

KAIZEN MATHS

Exit ticket

Indices and Exponent Laws

Fundamentals of Index Notation

Name _____

E1

Write in index form: $5 \times 5 \times 5 \times 5 \times 5$

E2

Evaluate: 4^3

E3

Express 48 as a product of prime factors in index form

ALG-934A-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Indices and Exponent Laws

Fundamentals of Index Notation

Name _____

E1

Write in index form: $5 \times 5 \times 5 \times 5 \times 5$

E2

Evaluate: 4^3

E3

Express 48 as a product of prime factors in index form

ALG-934A-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Indices and Exponent Laws

Fundamentals of Index Notation

Name _____

E1

Write in index form: $5 \times 5 \times 5 \times 5 \times 5$

E2

Evaluate: 4^3

E3

Express 48 as a product of prime factors in index form

ALG-934A-L1 · E

Show your working

Teacher answers & working

Indices and Exponent Laws · Fundamentals of Index Notation

ALG-934A-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Write in index form: $3 \times 3 \times 3$

Answer:

3^3

Working:

Identify the repeated factor.

The base is 3 and it appears 3 times.

Write the repeated multiplication as a power.

$3 \times 3 \times 3 = 3^3$

S2

Evaluate: 4^2

Answer:

16

Working:

4^2 means 4 multiplied by itself 2 times.

$4^2 = 4 \times 4$

$4 \times 4 = 16$

S3

Express 24 as a product of prime factors in index form

Answer:

$2^3 \times 3$

Working:

Divide 24 by prime factors systematically.

2 appears 3 times.

3 appears 1 time.

$24 = 2^3 \times 3$

S4

Simplify: a^{-1}

Answer:

$\frac{1}{a^1}$

Working:

A negative index means reciprocal, not a negative value.

$a^{-n} = \frac{1}{a^n}$

$a^{-1} = \frac{1}{a^1}$

Teacher answers & working

Indices and Exponent Laws · Fundamentals of Index Notation

ALG-934A-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

Write in index form: $5 \times 5 \times 5 \times 5 \times 5$

Answer:

5^5

Working:

Identify the repeated factor.

The base is 5 and it appears 5 times.

Write the repeated multiplication as a power.

$5 \times 5 \times 5 \times 5 \times 5 = 5^5$

E2

Evaluate: 4^3

Answer:

64

Working:

4^3 means 4 multiplied by itself 3 times.

$4^3 = 4 \times 4 \times 4$

$4 \times 4 \times 4 = 64$

E3

Express 48 as a product of prime factors in index form

Answer:

$2^4 \times 3$

Working:

Divide 48 by prime factors systematically.

2 appears 4 times.

3 appears 1 time.

$48 = 2^4 \times 3$